2. a)

i = the index of Vendor, $I = \{1, 2, 3\}$

Decision :

$$y_i = \begin{cases} 1, & \text{the university purchases computers from Vendor } i \\ 0, & \text{otherwise} \end{cases} \quad \in I$$

x_i = the number of computers purchased from Vendor i

Data :

p_i = unit price of computers purchased from Vendor i $\in I$

d_i = the total delivery charge if the university purchases computers from Vendor $i \in I$

m_i = the minimum quantity of computers required to order from Vendor $i \in I$

M_i = the maximum quantity of computers required to order from Vendor $i \in I$

N = the number of computers that the university need to purchase in total

Objective function:

$$\text{cost} = \sum_{i \in I} (\text{the cost} + \text{delivery charge}) \text{ from Vendor } i$$

$$= \sum_{i \in I} y_i (p_i x_i + d_i)$$

Constraints:

$$\sum_{i \in I} x_i = N$$

$$x_i > m_i y_i, \ \forall i \in I$$

$$x_i < M_i y_i, \ \forall i \in I$$

$$x_i \geq 0 \text{ and integer}, \ \forall i \in I$$

$$y_i \text{ binary}, \ \forall i \in I$$

Variable definitions:

x_i, y_i $\quad i \in I$

Model:

$$\min \sum_{i \in I} y_i (p_i x_i + d_i)$$

$$\text{s.t.} \sum_{i \in I} x_i = N \qquad \text{total computer numbers constraint}$$

$$\left.\begin{aligned} x_i &> m_i y_i, \ \forall i \in I \\ x_i &< M_i y_i, \ \forall i \in I \end{aligned}\right\} \text{order number of each vendor constraint}$$

$$x_i \geq 0 \text{ and integer}, \ \forall i \in I$$

$$y_i \text{ binary}, \ \forall i \in I$$